

Hepatic phenotypes of *EFL1*-related Shwachman-Diamond syndrome in a biopsy-validated study

To the Editor:

Shwachman Diamond syndrome (SDS), an inherited bone marrow failure syndrome, is characterized by exocrine pancreatic deficiency, predisposition to myeloid malignancies, and skeletal dysplasia. Other symptoms also include hepatic and renal involvement. Hepatomegaly and aminotransaminase elevation can be observed at an early age in patients with SDS, typically with an overall good prognosis. A follow-up with mild cholestasis may reflect primary alterations in liver metabolism in SDS. During longer follow-up, no major liver abnormalities are expected.¹ Biallelic pathogenic variants in the *SBDS* gene account for the majority of SDS cases.^{2,3} Recently, biallelic mutations in the *DNAJC21* gene and *EFL1* gene and *de novo* dominant mutations of the *SRP54* gene have been linked to SDS or SDS-like phenotypes.^{4,5} The *EFL1* gene, located on chromosome 15q25.2, encodes the elongation factor-like GTPase 1 protein.⁶ Currently, only 13 mutations have been reported in 18 patients worldwide (Table S1, Fig. S1). In China, no SDS cases with *EFL1* gene mutations have been recorded. Here we report a Chinese female patient with two novel *EFL1* mutations, who exhibited cirrhosis and thrombocytopenia and was diagnosed with SDS.

A 44-year-old Chinese woman was admitted to our hospital in June 2020. She had a 1-month history of abdominal distension and lower-limb edema, as well as a 14-year history of mild thrombocytopenia. She had a lengthy history of moderate diarrhea, but no history of liver disease, alcohol or tobacco use. She exhibited malnourishment (BMI 16.5) and pectus carinatum and splenomegaly on physical examination. Following clinical examinations, whole-exome sequencing (WES) was used to identify the genetic results (Fig. 1A).

Leukopenia and thrombocytopenia were observed in her initial hematologic examination (white blood cell [WBC]: $2.6 \times 10^9/L$; platelet [PLT]: $37 \times 10^9/L$), and subsequently progressed into severe pancytopenia (hemoglobin: 89 g/L; WBC: $0.94 \times 10^9/L$; PLT: $29 \times 10^9/L$). Additionally, she showed elevated liver enzymes (alanine aminotransferase: 44 U/L; aspartate aminotransferase: 42 U/L) and decreased albumin (30 g/L), cholinesterase (3,424 U/L), and prothrombin time activity (48.7%). Pancreatic function tests revealed low serum amylase (8 U/L) and low stool fecal elastase.

The abdominal MRI showed diffuse liver damage, cirrhosis, splenomegaly, mild ascites, splenic vein tortuosity, splenic vein thrombosis, and abnormal perfusion signals in the liver during the arterial phase. Cholecystitis and an indistinct pancreatic margin were also detected (Fig. 1C). The liver stiffness measurement was 14.7 kPa, indicating a

significant degree of liver fibrosis. Other possible causes of liver-related phenotypes were excluded through clinical examinations (Table S2).

Bone marrow puncture revealed increased production of granulocyte and erythrocyte series and a significantly elevated megakaryocyte count and thrombocytopenia. Hepatic venous pressure gradient (30.4 cmH₂O) was calculated from a transjugular liver biopsy. Histopathological examination showed hepatic parenchymal separation by fibrosis septa and disordered hepatocyte plate arrangement (Fig. 1E). Immunohistochemical analysis was consistent with G1S3 (inflammation level 1, and liver fibrosis level 3), suggesting atypical liver fibrosis.

Two heterozygous novel variants in the *EFL1* gene (NM_024580.6), the c.1581_1587delCAAATAC (p.K528Vfs*11) and c.74T>C (p.L25S) mutations, were identified. Sanger sequencing confirmed that two mutations came from her healthy mother and father, respectively (Fig. 1B,D). These two variants were predicted to be deleterious by multiple *in silico* tools and classified as pathogenic and likely pathogenic, respectively (Table S3).

The frameshift mutation p.K528Vfs*11 is located in the crucial functional dynamic domain II, introducing a premature termination codon and resulting in the loss of part of domain II, as well as the domains III, IV, and V (Fig. S1). The III/IV domains of *EFL1* bind to its cofactor *SBDS*. III/IV domains resulting in the establishment of a low-affinity binding mode on the ribosome. They facilitate the dissociation of eIF6 by further rotational displacement of *SBDS* and an *SBDS*-dependent conformational change of *EFL1*.⁷ Loss of these domains may impede the allosteric activation of *EFL1* and hinder the eviction of eIF6 from the nascent 60S. Domain V acts in mediating *SBDS*'s repositioning around helix 69.⁸ Protein structure change analysis of mutation p.L25S indicates that the mutation results in a clash (Fig. S2A,B) that may compromise the stability of the protein structure. In addition, this mutation impacts the amino acids that are highly conserved across species (Fig. S2C). Thus, we concluded that the p.K528Vfs*11 and p.L25S mutations hinder the function of *EFL1* protein and validate the diagnosis of SDS resulting from *EFL1* gene mutations.

While liver symptoms are typically mild in most patients with SDS, we report on a Chinese patient with severe cirrhosis symptoms related to SDS for the first time. After treatment, the patient's condition remained stable until follow-up in October 2021. This case suggests that we should reconsider the severity of the liver phenotype related to SDS. Genetic testing is crucial for the diagnosis of SDS.

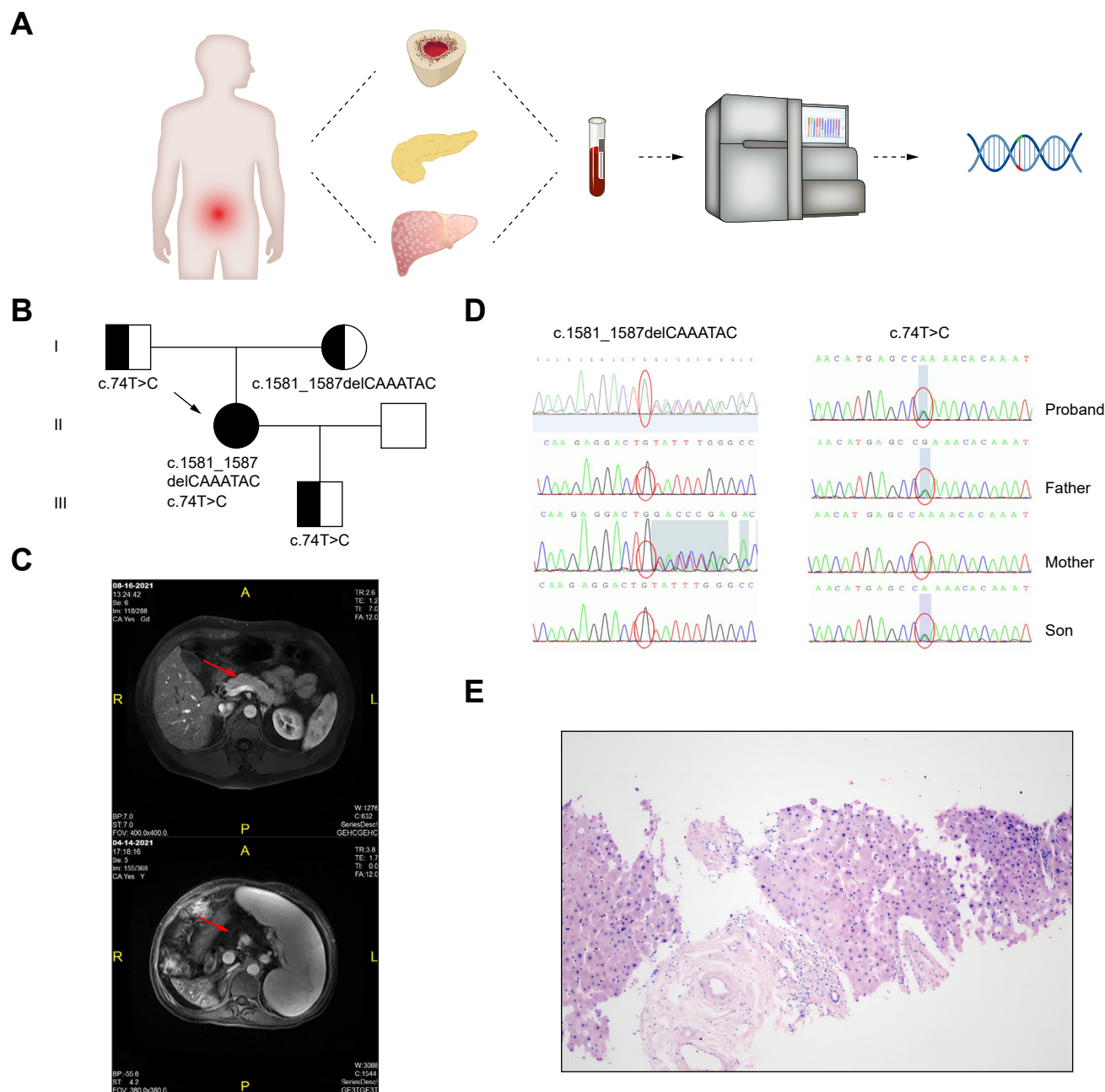


Fig. 1. Genotypic and phenotypic characteristics of the *EFL1* patient. (A) Affected organs of the patient and disease diagnosis process. (B) Pedigree chart of the family in three generations. Arrow indicated the proband. Her parents and son are healthy carriers. (C) MRI image of the abdomen (portal phase; axial view). Arrow points to the pancreas of a normal person (upper image), with intact architecture; and the patient (lower image), with indistinct border and peripancreatic fatty infiltration. (D) Sanger sequencing of the two *EFL1* mutations. The *EFL1* c.1581_1587delCAAATAC mutation at position chr15:82512017 and c.74T>C mutation at position chr15:82554046 from her mother and father, respectively. Shown with antisense sequence. (E) The pathological findings of liver tissues via H&E staining (x100). Histopathological analysis shows the enlargement of portal area, hyperplasia of fibrous tissue and formation of fibrous septa. Mild inflammatory cell infiltration was present, with no inflammatory cells at the interface region.

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Conflict of interest

All authors declare that there is no potential conflict of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

BZ designed and performed the study. YG performed data analysis. SL collected and interpreted the clinical data. BZ, YG, and XL wrote the manuscript. SY conceived and supervised the research. All authors reviewed the manuscript.

Ethics approval statement

This study was conducted in accordance with the principles of the Declaration of Helsinki and approved by the Institutional Review Board of the Fifth Medical Center of Chinese PLA General Hospital.

Patient consent statement

Written informed consent was obtained from the patient for gene sequencing and the publication of the case report.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.04.030>.

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